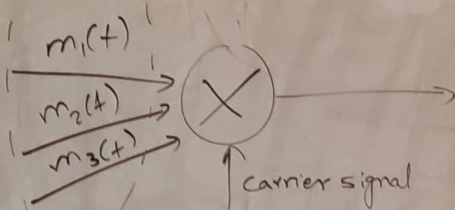


multi information
Signal



* $m_1(t) = A m_1 \cos(2\pi f_{m1} t)$ { more than one }
 * $m_2(t) = A m_2 \cos(2\pi f_{m2} t)$ { Frequency }

General form

$S(t) = A_c (1 + \mu \cos(2\pi f_m t)) \cos(2\pi f_c t)$ (single tone)
 (AM-DSB-R) modulated signal

$\mu = \frac{A_m}{A_c}$
 $S_1(t) + S_2(t)$

Lec. 3

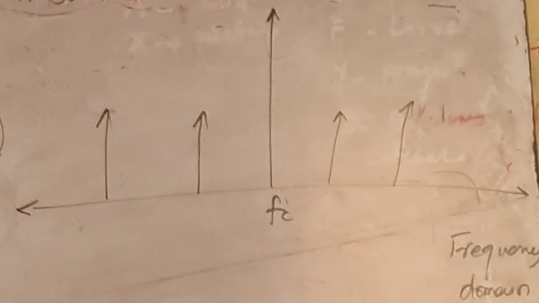
AM-FC

Single tone vs Multitone

AM

FC
DSB-FC

* Bandwidth = $f_{max} - f_{min} = 2 f_{max}$



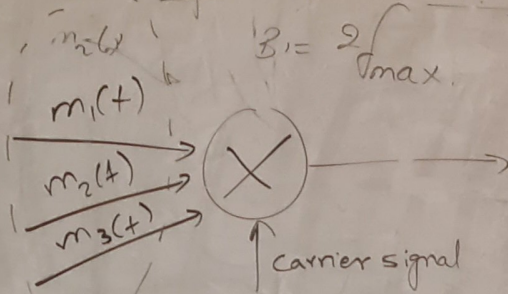
For multi tone

$m(t) = m_1(t) + m_2(t)$

$S(t) = A_c \left[(1 + \mu_1 \cos(2\pi f_{m1} t)) + (1 + \mu_2 \cos(2\pi f_{m2} t)) \right] \cos(2\pi f_c t)$

* $S(t) = A_c \cos(2\pi f_c t) + \frac{\mu_1 A_c}{2} \cos(2\pi(f_c - f_{m1})t) + \frac{\mu_1 A_c}{2} \cos(2\pi(f_c + f_{m1})t) + \frac{\mu_2 A_c}{2} \cos(2\pi(f_c - f_{m2})t) + \frac{\mu_2 A_c}{2} \cos(2\pi(f_c + f_{m2})t)$

multi information
Signal



* $m_1(t) = A m_1 \cos(2\pi f_{m1} t)$ { more than one }
 * $m_2(t) = A m_2 \cos(2\pi f_{m2} t)$ { Frequency }

General form

$S(t) = A_c (1 + M \cos 2\pi f_m t)$ single tone
(AM-DSB-FC)
 modulated signal

$S_1(t) + S_2(t)$ $\mu = \frac{A_{m1}}{A_c}$

Lec. 3

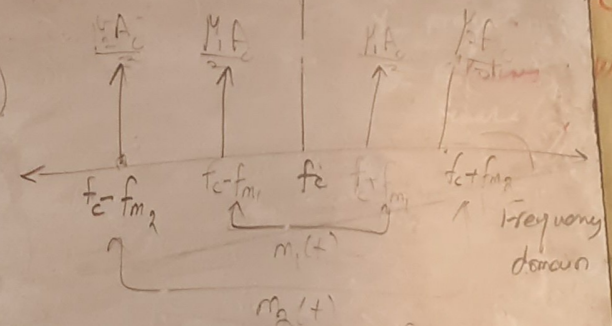
AM-FC

Single tone vs Multi tone

AM

FC
DSB-FC

* Bandwidth = $f_{max} - f_{min} = 2 f_{max}$



For multi tone

$m(t) = m_1(t) + m_2(t)$

BW = $f_{max} - f_{min} = 2 f_{max}$

$S(t) = A_c [1 + M_1 \cos 2\pi f_{m1} t] + [1 + M_2 \cos 2\pi f_{m2} t] \cos 2\pi f_c t$

* $S(t)_{multi} = A_c \cos 2\pi f_c t + \frac{M_1 A_c}{2} \cos(2\pi(f_c - f_{m1})t) + \frac{M_1 A_c}{2} \cos(2\pi(f_c + f_{m1})t) + \frac{M_2 A_c}{2} \cos(2\pi(f_c - f_{m2})t) + \frac{M_2 A_c}{2} \cos(2\pi(f_c + f_{m2})t)$

multi information
Signal

$$M_t = \sqrt{M_1^2 + M_2^2 + \dots + M_n^2}$$

$$P_t = P_c \left(1 + \frac{M_t^2}{2} \right)$$

$$P_t = P_c + P_{usb_1} + P_{usb_2} + P_{lsb_1} + P_{lsb_2}$$

$$\eta = \frac{M_t}{2}$$

Lec. 3

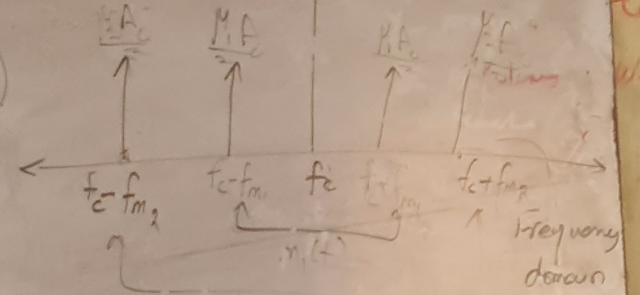
AM-FC

Single tone vs Multitone

AM

$$* \text{Bandwidth} = f_{\max} - f_{\min} = 2f_{\max}$$

FC
DSB-FC



For multi tone

$$m(t) = m_1(t) + m_2(t)$$

$$Bw = f_{\max} - f_{\min} = 2f_{\max}$$

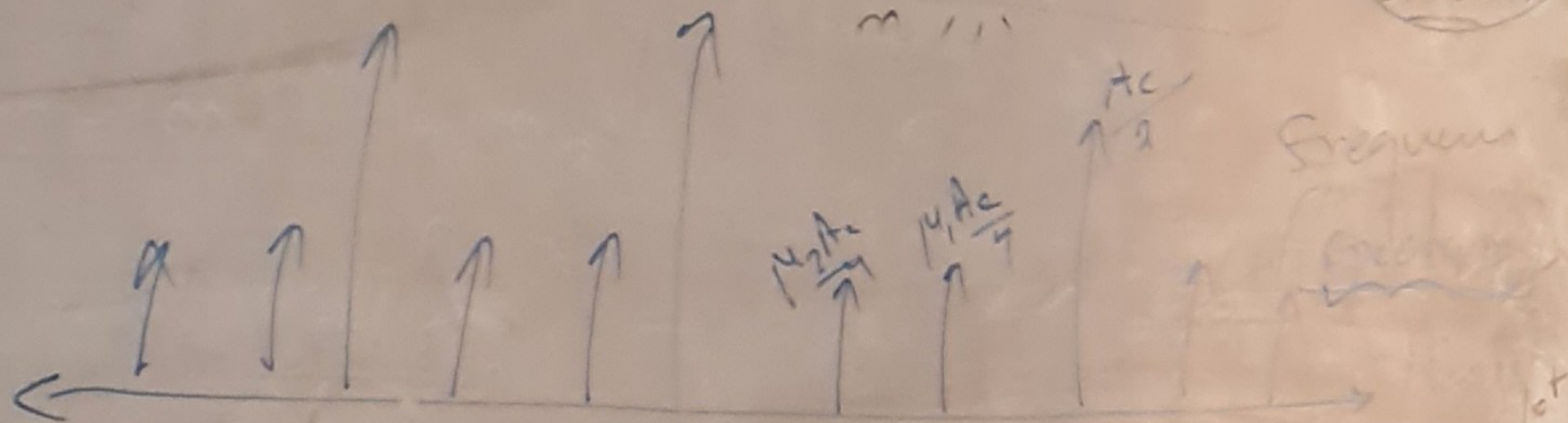
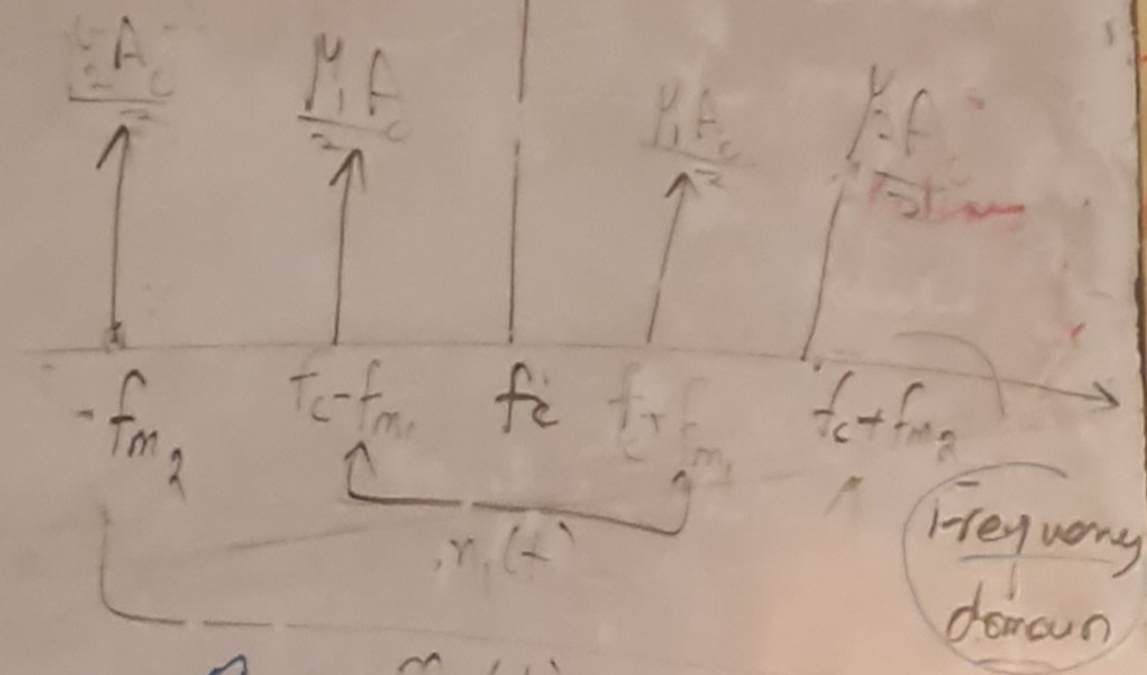
$$S(t) = A_c \left[1 + M_1 \cos(2\pi f_{m1} t) \right] + \left[1 + M_2 \cos(2\pi f_{m2} t) \right] \cos(2\pi f_c t)$$

$$* S(t)_{\text{multi}} = A_c \cos(2\pi f_c t) + \frac{M_1 A_c}{2} \cos(2\pi(f_c - f_{m1})t) + \frac{M_1 A_c}{2} \cos(2\pi(f_c + f_{m1})t) + \frac{M_2 A_c}{2} \cos(2\pi(f_c - f_{m2})t) + \frac{M_2 A_c}{2} \cos(2\pi(f_c + f_{m2})t)$$

AM

* Bandwidth = $f_{\max} - f_{\min} = 2f_{\max}$

FC
DSB-FC



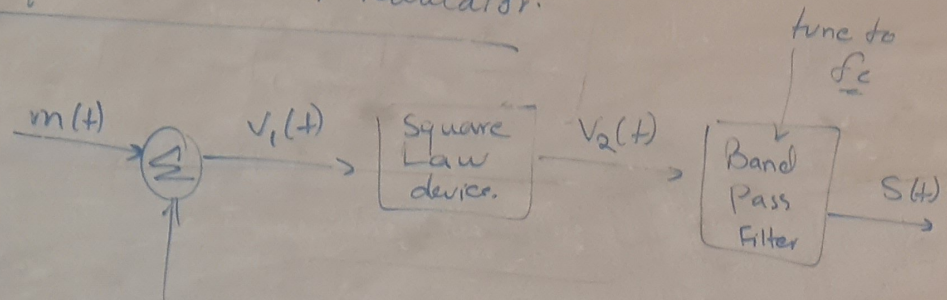
Lec. 3

AM-FC

AM

FC
DSB-FC

Square Law Modulator.



$$A_c \cos(2\pi f_c t)$$

$$V_1(t) = m(t) + A_c \cos(2\pi f_c t), \quad V_2(t) = K_1 V_1(t) + K_2 V_1^2(t) \Rightarrow V_2(t) = K_1 m(t) + K_2 m^2(t) + K_2 A_c^2 \cos^2(2\pi f_c t) + K_1 A_c \left[1 + \frac{2K_2}{K_1} m(t) \cos(2\pi f_c t) \right]$$

$$S(t) = A_c [1 + K_a m(t)] \cos 2\pi f_c t$$

Lec. 3 AM-FC

AM

FC
DSB-FC

(gain/Loss)

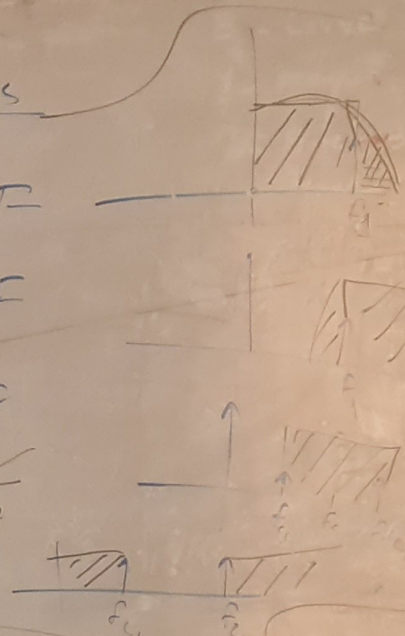
Filters

LPE

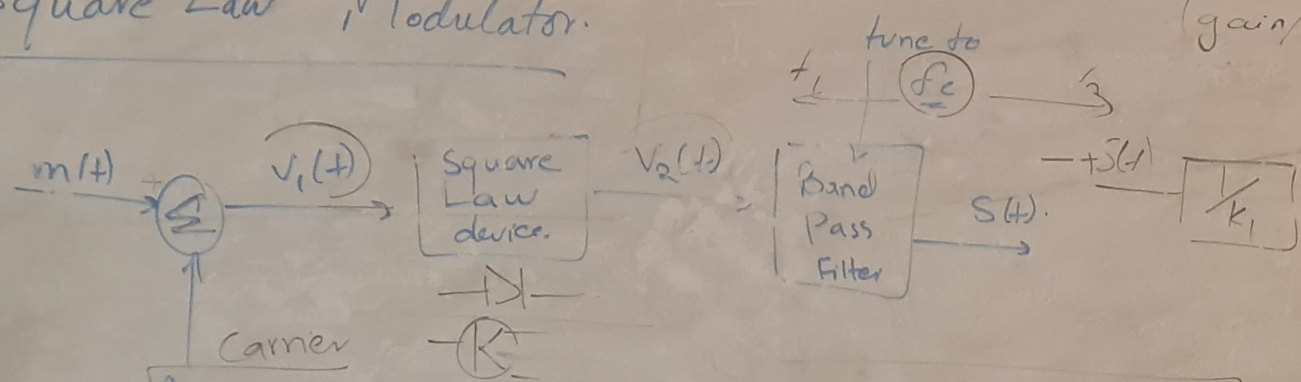
HPE

BPF

BSF



* Square Law Modulator.



$$V_1(t) = m(t) + A_c \cos(2\pi f_c t)$$

$$V_2(t) = K_1 V_1(t) + K_2 V_1^2(t)$$

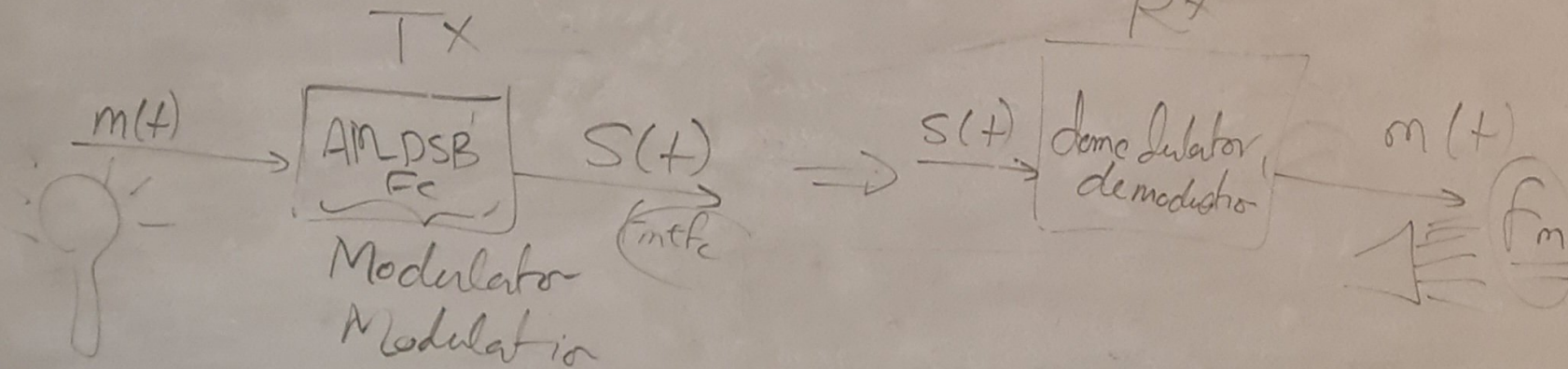
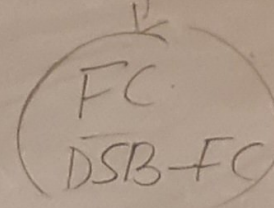
$$S(t) = A_c [1 + K_a m(t)] \cos 2\pi f_c t$$

$$V_2(t) = K_1 m(t) + K_2 m^2(t) + K_2 A_c^2 \cos^2(2\pi f_c t) + K_1 A_c [1 + \frac{K_2}{K_1} m(t) \cos 2\pi f_c t]$$

Lec. 3

AM-FC

AM



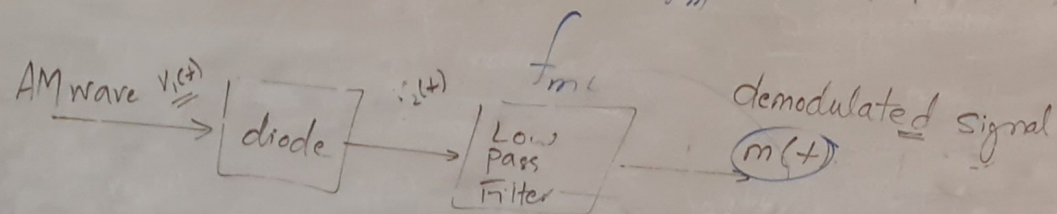
Lec. 3

AM-FC

AM

FC
DSB-FC

Envelope detector.

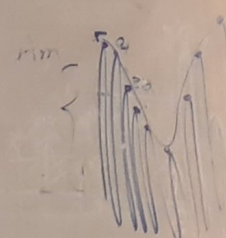
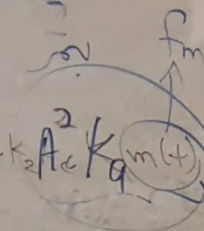


$$S(t) = A_c [1 + k_a m(t)] \cos 2\pi f_c t$$

$$V_2(t) = k_1 V_1(t) + k_2 V_1^2(t)$$

$$V_2(t) = k_1 A_c \cos 2\pi f_c t + k_1 A_c k_a m(t) \cos 2\pi f_c t + \dots + k_2 A_c^2 k_a^2 m^2(t) \cos 2\pi f_c t + k_2 A_c^2 k_a m(t) \cos 2\pi f_c t$$

Gain Loss



Lec. 3 AM-FC

An audio wave = $\left[0.2 \cos 2(1000 \pi t) + 0.1 \cos 7(1000 \pi t) \right]$

is modulated with a carrier signal = $\left[4 \cos(1000 \times 10^3 \pi t) \right]$

Find:

① Sketch the complete Frequency domain wave.

② Find total power

P_t

③ Bandwidth

BW

④ Sideband power only

P_{SB}

⑤ Efficiency